UNIT 1 – Data Analysis with R

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Subject: Data Analysis by R Programming

Assignment 1: Basics of Data Analysis
Definition: Data analysis means examining data to find useful information and patterns.
Importance: Helps in better decision making and improving business performance.
Steps: Data Collection → Cleaning → Analysis → Visualization → Decision Making
Draw Flowchart in notebook.

Assignment 2: Data Types
Age – Numeric
Gender – Categorical
Salary – Numeric
Pass/Fail – Logical
Temperature – Numeric
Laptop (R Code):

```
age <- 20
gender <- "Male"
salary <- 50000
result <- TRUE
temp <- 36.5
```

Assignment 3: R Environment
R executes code line by line and shows output in console.
Explain RStudio interface.
Laptop (R Code):

```
print("Hello R")
a <- 10
b <- 5
print(a+b)
```

```
a <- 10
b <- 5

a+b
a-b
a*b
a/b
a%%b

# function
percentage <- function(marks, total){
  (marks/total)*100
}

# pass fail
marks <- 40
if(marks >= 35){
  print("Pass")
}else{
  print("Fail")
```

```r
# pass fail
marks <- 40
if(marks >= 35){
  print("Pass")
}else{
  print("Fail")
}

# built-in
data <- c(10,20,30)
mean(data)
sum(data)
length(data)
```

```
> a <- 10
> b <- 5
>
> a+b
[1] 15
> a-b
[1] 5
> a*b
[1] 50
> a/b
[1] 2
> a%%b
[1] 0
>
```

```
> # function
> percentage <- function(marks, total){
+   (marks/total)*100
+ }
>
> # pass fail
> marks <- 40
> if(marks >= 35){
+   print("Pass")
+ }else{
+   print("Fail")
+ }
[1] "Pass"
 - -
>
> # built-in
> data <- c(10,20,30)
> mean(data)
[1] 20
> sum(data)
[1] 60
> length(data)
[1] 3
>
```

RStudio

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Lecture 2.R | Lecture 3.R | Untitled1* | Lecture 1.R

```r
## Assignment 1

print("Hello R")
a <- 10
b <- 5
print(a+b)
```

Console | Terminal | Background Jobs

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> print("Hello R")
[1] "Hello R"
> a <- 10
> b <- 5
> print(a+b)
[1] 15
> N
```

Environment | History | Connections | Tutorial

Data

list	List of 6
list1	List of 3
list2	List of 3
list3	List of 6
thelist	List of 5
twolist	List of 4

Values

a	10
b	5
fruits	chr [1:3] "apple" "banana" "mango"

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R: Matrices

matrix {base} R Documentation

Matrices

Description

`matrix` creates a matrix from the given set of values.

`as.matrix` attempts to turn its argument into a matrix.

`is.matrix` tests if its argument is a (strict) matrix.

Usage

```r
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)

as.matrix(x, ...)
```

RStudio

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Lecture 2.R | Lecture 3.R | Untitled1* | Lecture 1.R

```r
## Assignment 1

age <- 20
gender <- "Male"
salary <- 50000
result <- TRUE
category <- "Electronics"
temp <- 36.5
active <- TRUE
```

Console | Terminal | Background Jobs

```
> age <- 20
> gender <- "Male"
> salary <- 50000
> result <- TRUE
> category <- "Electronics"
> temp <- 36.5
> active <- TRUE
>
```

Environment | History | Connections | Tutorial

Values

active	TRUE
age	20
category	"Electronics"
gender	"Male"
result	TRUE
salary	50000
temp	36.5

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R: Matrices

matrix {base} R Documentation

Matrices

Description

`matrix` creates a matrix from the given set of values.

`as.matrix` attempts to turn its argument into a matrix.

`is.matrix` tests if its argument is a (strict) matrix.

Usage

```r
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)

as.matrix(x, ...)
```

RStudio

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Lecture 2.R | Lecture 3.R | Untitled1* | Lecture 1.R

```r
## Assignment 1

age <- 20
gender <- "Male"
salary <- 50000
result <- TRUE
category <- "Electronics"
temp <- 36.5
active <- TRUE
```

9:15 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.5.1 · ~/

```
> age <- 20
> gender <- "Male"
> salary <- 50000
> result <- TRUE
> category <- "Electronics"
> temp <- 36.5
> active <- TRUE
>
```

Environment | History | Connections | Tutorial

Global Environment

Values

active	TRUE
age	20
category	"Electronics"
gender	"Male"
result	TRUE
salary	50000
temp	36.5

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R: Matrices

matrix {base} R Documentation

Matrices

Description

`matrix` creates a matrix from the given set of values.

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Usage

```r
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)

as.matrix(x, ...)
```

RStudio

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Lecture 2.R | Lecture 3.R | Untitled1* | Lecture 1.R

```r
## Assignment 1

# vector
marks <- c(80, 75, 90, 85)
sum(marks)
mean(marks)

# matrix
mat <- matrix(1:9, nrow=3)
sum(mat)

# list
student <- list(name="Rahul", age=20, marks=85)

# data frame
df <- data.frame(Name=c("A","B"), Marks=c(80,90))
df$Grade <- c("A","A")
```

17:23 (Top Level) R Script

Console | Terminal | Background Jobs

R 4.5.1 · ~/

```
> # vector
> marks <- c(80, 75, 90, 85)
> sum(marks)
[1] 330
> mean(marks)
[1] 82.5
>
> # matrix
> mat <- matrix(1:9, nrow=3)
> sum(mat)
[1] 45
>
> # list
> student <- list(name="Rahul", age=20, marks=85)
>
```

Environment | History | Connections | Tutorial

Global Environment

Data

df	2 obs. of 3 variables
mat	int [1:3, 1:3] 1 2 3 4 5 6 7 8 9
student	List of 3

Values

marks	num [1:4] 80 75 90 85

Files | Plots | Packages | Help | Viewer | Presentation

R: Matrices

matrix {base} R Documentation

Matrices

Description

`matrix` creates a matrix from the given set of values.

`as.matrix` attempts to turn its argument into a matrix.

`is.matrix` tests if its argument is a (strict) matrix.

Usage

```r
matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,
       dimnames = NULL)

as.matrix(x, ...)
```

Assignment 4: Data Structures
Vector: same type data
Matrix: 2D data
List: mixed data
Data Frame: table format
Laptop (R Code):

```
marks <- c(80,75,90)
sum(marks)
mean(marks)

mat <- matrix(1:9,3)
sum(mat)
```

Assignment 5: Functions
Write in Journal:
Functions help reuse code.
Logical conditions help decision making.
Laptop (R Code):

```
percentage <- function(m,t){
 (m/t)*100
}

percentage(450,500)
```